Optiver Trading Challenge

Trading Challenge: ETF-Future quoting strategy

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1 Introduction

In this trading challenge you will work on an algorithm to quote an index ETF based on the value of a futures contract on the same underlying index.

The challenge builds upon concepts covered in the Dual Listing quoting strategy. Again, your algorithm will take prices from a liquid instrument (the index future) to a less liquid instrument (the ETF) and hedge using the liquid instrument. The difference is that the prices between the two traded instruments no longer relate on a one-to-one basis.

After finishing it, your algorithm will continuously and profitably provide liquidity in the ETF through inserting limit orders based on the prices of a Futures, ensuring to keep total outstanding positions hedged.

2 Trading Challenge

The exchange lists five stocks which together form an index (the OB5X). This index cannot be traded directly, but there are several different products based on it: derivatives like the futures contracts, and an ETF. Refer to the Index-Derived Instruments section of the Optibook Guide for a comprehensive overview on how these relate.

As the instruments all relate back to the same underlying quantity, their prices should relate in a predictable manner. Analogously to the Dual Listing exercise, this price relationship only holds in the long-run equilibrium. On shorter time-scales, supply and demand and a variation of market participants may push the prices in the order books out of line with each other. Your quoting algorithm will assist in bridging that information gap, by providing prices in the Index ETF that arise from the current prices of the Index Futures contract.

Implement the following trading strategy based on the provided sample code in the etf directory on Cloud9/Jupyter (a hint: make a copy of the sample algorithm before modifying it, for future reference).

2.1 Quoting Strategy

Trading Challenge:

- Implement a profitable quoting strategy of the Index ETF based on the prices of one of the Index Futures; ensuring to stay hedged as much as possible

Use your dual listing quoting algorithm as reference. The quoting and hedging strategy is the same, so you can likely follow a similar structure. It is the relationship between the instruments which is more generalized.

There are multiple index futures listed on the exchange, select one to use as a “base contract”, that is, the one you will use to trade the ETF. For example, pick the one with the closest expiry. Ignore the other futures for now.

Your algorithm should wake up periodically and, on each iteration, take the following actions:

- Remove current remaining outstanding limit orders in the ETF, if any.

- Convert the Future bid and ask prices into ETF bid and ask prices by using the applicable pricing formulas * from the Instrument Reference.

  Hint: use the pricing formula for the futures to solve for the index value X, then insert the index value into the ETF pricing formula to determine the ETF value. You can repeat this calculation for the bid and ask side to arrive at a fair bid/ask price for the ETF.

- Subtract a credit from the fair ETF bid price and add a credit to the fair ETF ask price. These are the profit margins you require.

- Round the prices with credit to the nearest tick, ensuring to round bid prices down and ask prices up (why?). These are the prices you wish to quote the ETF at.

- Determine the volumes you are willing to quote; they may be different for the bid and ask side. Take into account the exchange position limits and how much volume you can hedge in the futures based on the current order book.

- Insert new limit orders in the ETF for the calculated prices and volumes.

- Wait some time for your limit orders to stay in the order book; both sides may trade, either partially or fully or not at all.

- Determine your current position in both the ETF and the Futures. What is your exposure to movement in the underlying index (take the derivative with respect to X). How many futures would you need to trade to hedge that exposure?

- Insert an IOC against the futures top level to hedge your full portfolio against movement in the index; using the volume you calculated.

- (Repeat from step 1)

* The applicable interest rate for futures pricing is $\mathbf{3\% = 0.03}$. The time to expiry can be calculated using the calculate_time_to_date() function in common/libs.py. The exact expiry timestamp is a property of the Instrument object corresponding to the Futures instrument.

You may deviate from the above step-by-step outline wherever you wish, as long as the core concepts, such as hedging, credit and determining the correct volumes are correctly considered.

2.2 (Optional) Expand to more index products

This exercise is optional, and likely to be more challenging, depending on how you set up your previous solutions.

The OB5X index has a relationship to the ETF and to 3 individual futures, on top of that the index may be replicated using the individual constituent stocks. As such there are 5 different ways to arrive at a currently correct index price. Due to supply and demand, these different calculated index values will not always align.

- Can you find a way to make use of (more of) the price discrepancies than the one between the Future-ETF pair of 2.1.1.? For example, how can you make use of the other two futures?

- Can the constituents-based index value come into this?